Late deliveries drive 1–2 star reviews

SQL + Power BI case study on real public data · Olist Brazilian e-commerce (Kaggle) · 96,470 delivered orders, Sep 2016 – Aug 2018 · Ahmed Hatem Dahi

6.8% of orders late | review **4.21 → 1.69** when 8+ days late

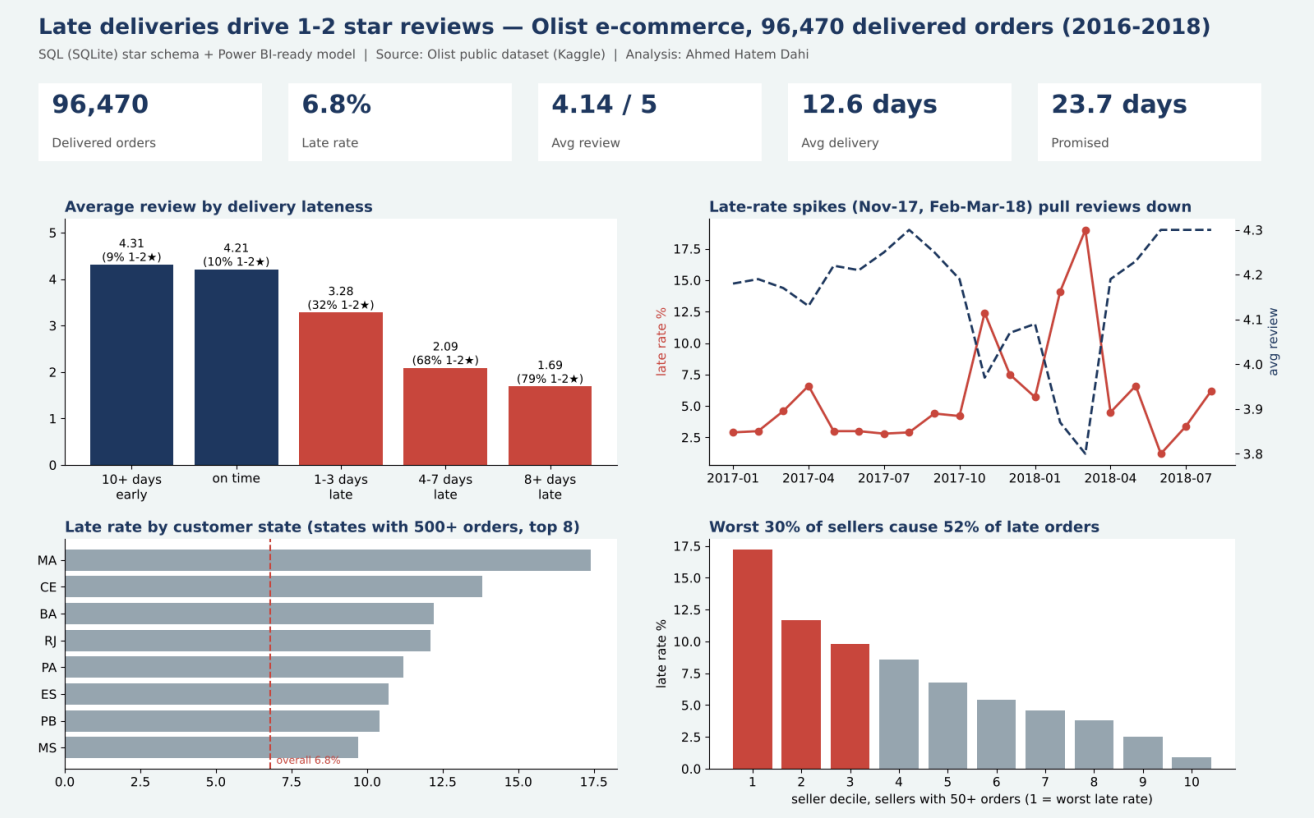
52% of late orders from 30% of sellers | late rate **19%** in Mar-2018

Business question

What drives bad reviews on a marketplace, and where should operations act first?

Data and method

Loaded 8 raw tables (99,441 orders, 112,650 items, 100,000 reviews, 3,095 sellers) into SQLite and built a star schema in SQL: `fact_orders` (one row per delivered order with delivery days, promised days, days late, review), `fact_order_items`, `dim_category`, `dim_customer_state`. Eight documented queries use CTEs and window functions (`RANK`, `NTILE`, `ROW_NUMBER`, `SUM() OVER`). The same tables are exported as CSV with a DAX measure file so the model opens in Power BI Desktop.



Findings

#	Finding	Evidence (query)
1	Reviews collapse once an order is late: average 4.21 on time, 3.28 at 1–3 days late, 2.09 at 4–7 days, 1.69 at 8+ days. Share of 1–2 star	03_review_by_lateness

	reviews goes from 10% to 79%.	
2	Lateness is concentrated: among sellers with 50+ orders, the worst 30% (129 sellers) produce 52% of late orders; the worst decile is late 17.2% of the time vs 0.9% for the best.	07_seller_concentration
3	Late rate spikes after demand peaks: 12.4% in Nov-2017 (Black Friday) and 14.1% / 19.0% in Feb / Mar-2018, when the average review fell to 3.8.	05_monthly_trend
4	North-east states are late most often (MA 17.4%, CE 13.8%, BA 12.2%) and pay the most freight (MA 26% of basket value vs SP 14%).	04, 09
5	Customers whose first order was late came back slightly less (2.5% vs 3.0% repeat rate); repeat purchase is low overall, so this is a weak signal.	08_repeat_customers

Recommendations

1. Seller scorecard: flag sellers in the worst late-rate decile weekly and require a shipping-time fix or a longer promised date.
2. Peak-season plan: extend promised dates and add carrier capacity before Black Friday and February.
3. Regional promise: set promised dates by state for the north-east instead of one national estimate.

Example query

```
WITH s AS (SELECT i.seller_id, COUNT(DISTINCT f.order_id) orders, SUM(f.is_late)
late_orders, AVG(f.is_late) late_rate
           FROM fact_order_items i JOIN fact_orders f ON f.order_id = i.order_id
           GROUP BY i.seller_id HAVING COUNT(DISTINCT f.order_id) >= 50),
ranked AS (SELECT *, NTILE(10) OVER (ORDER BY late_rate DESC) decile FROM s)
SELECT decile, COUNT(*) sellers, SUM(orders) orders, SUM(late_orders) late_orders,
       ROUND(100.0*SUM(late_orders)/SUM(orders),1) late_rate_pct
FROM ranked GROUP BY decile ORDER BY decile;
```

Notes: an order counts as late when the delivery date is after the estimated date. Revenue = item prices in BRL (freight excluded). Independent study on public data, not client work. Data licence: CC BY-NC-SA 4.0.